

AI for Cybersecurity Week 2 Lesson 1: Build a Cyberattack Classifier

Activity Description

In the activity, you will build a Cyberattack Classifier using a cybersecurity dataset containing:

- network traffic records,
- login attempts,
- email characteristics,
- or malware features.

Use one of the two suggested datasets below, or one of the other publicly available datasets featured in the lesson:

- CICIDS2017: <https://www.kaggle.com/datasets/chethuhn/network-intrusion-dataset>
- Phishing Email Dataset: <https://www.kaggle.com/datasets/naserabdullahalam/phishing-email-dataset>

Task

Complete the following tasks in 2-3 pages:

Part A — Dataset Understanding

1. Identify the:
 - input features,
 - target labels,
 - and cybersecurity problem type.
2. Why is this a supervised learning problem?

Part B — Model Selection

3. Identify the most suitable algorithm, from the ones studied in the lesson, e.g. decision tree, logistic regression, etc.
4. Explain why your selected algorithm fits the dataset.
5. Write the Python code to read and classify the attack. Include a link to your Python code.

Part C — Cybersecurity Analysis

5. What cybersecurity threat is being classified?
6. Why is classification important in cybersecurity?
7. What problems could occur if the model misclassifies attacks?

Provide your solution in PDF format and include a public link to your Google Colab and dataset, so your tutor and colleagues can access it.